SIGNALLING & INTERLOCKING TECHNICAL SPECIFICATION

Electronic Interlocking (EI) for Major Station Yard

Specification No.	S&T/EI/SPEC/2026/017
Station	Sample Junction (SMJ) - 6 platforms, 42 points, 58 signals
Safety Integrity Level	SIL-4 (EN 50126 / 50128 / 50129)
Issued By	Signal & Telecom Engineering Dept. (fictional)

1. System Overview

The Electronic Interlocking shall control all points, signals, track circuits and axle counters within the station limits. The system shall employ a 2-out-of-2 (2oo2) vital processor architecture with hot-standby redundancy and shall conform to the Railway Board approved interlocking principles. All route setting shall be through VDU-based operator control with a fallback panel interface.

2. Functional Requirements

Req. ID	Requirement	Verification
FR-001	Route shall be set only if all points are proved locked in required position	Simulation + site test
FR-002	Conflicting routes shall never be set simultaneously	Interlocking table check
FR-003	Signal shall revert to danger within 1.0 s of track circuit occupation ahead	Timing test
FR-004	Route release timer for emergency cancellation: 120 s (main), 60 s (shunt)	Functional test
FR-005	Approach locking for main signals, with 180 s release	Functional test
FR-006	All events logged with 10 ms timestamp resolution, retained 90 days	Audit

3. Field Equipment Parameters

Equipment	Type	Rating / Setting	Standard
Signal lamp	LED, 3-aspect colour light	12 V DC, 10 W, life > 100,000 h	RDSO
Point machine	Electric, 3-phase non-trailable	110 V AC, throw 143 mm, 5 s operating time	RDSO
Track circuit	Audio Frequency, jointless	Frequencies 1.7 / 2.3 / 2.6 kHz, 20 m to 450 m	RDSO
Axle counter	Digital, single section	Wheel detector gap 5 mm min.	IRS S 100
Signalling cable	Quad armoured, 1.5 sqmm	Insulation > 20 MOhm/km at 500 V DC	IRS S 63
Power supply	Dual feed with SMPS	110 V DC, float 2.25 V/cell, 2 h battery backup	RDSO

4. Sample Route Interlocking Table

Route	From Signal	To Signal	Points Required	Tracks Clear	Conflicting Routes
R-01	S1	S5	P1(N), P3(N)	T1, T2, T3	R-04, R-07
R-02	S1	S6	P1(N), P3(R)	T1, T2, T4	R-05
R-03	S2	S7	P2(R), P5(N)	T5, T6	R-06, R-09
R-04	S3	S5	P3(N), P6(R)	T3, T7	R-01
R-05	S4	S6	P4(N)	T4, T8	R-02

N = Normal position, R = Reverse position.

5. Testing and Commissioning

Stage	Test	Acceptance Criterion
Factory	Software logic verification	100% of interlocking table matched
Site	Point detection and locking	Detection lost at 3.5 mm opening
Site	Track circuit shunt test	Shunt 0.5 ohm must drop relay
Site	Emergency route release	Release only after timer expiry
Final	Independent safety assessment	ISA certificate issued before commissioning

6. Reliability Targets

System availability shall be not less than 99.995%. Mean Time Between Wrong-Side Failures (MTBWSF) shall exceed 10^9 hours. Mean Time To Repair (MTTR) shall not exceed 2 hours for any replaceable module.